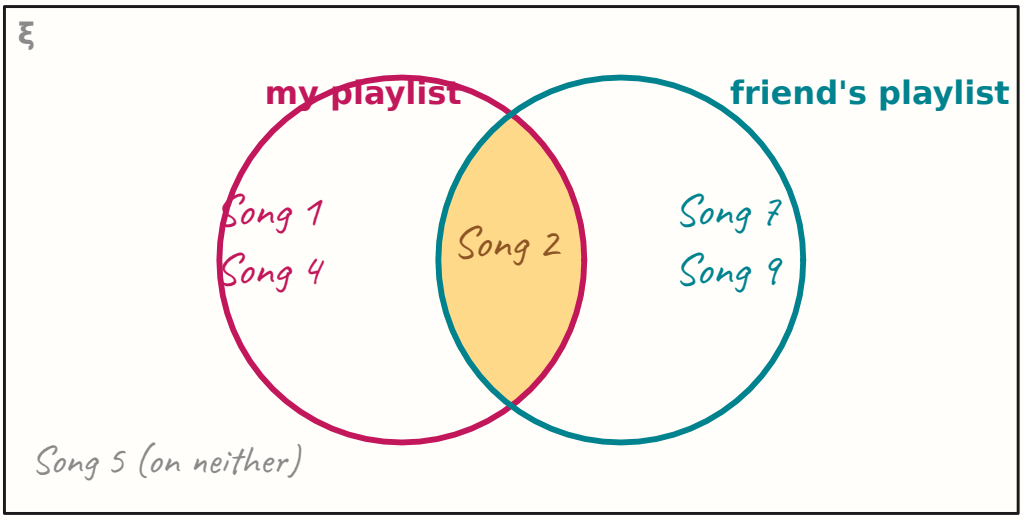
Chapter 6

Sets

collections of things, drawn as hula hoops on the floor



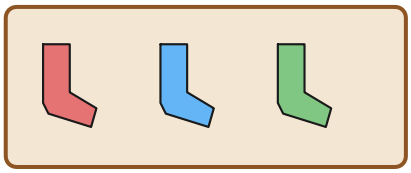
A **set** is just a collection of things. That's all. Your sock drawer is a set of socks; a playlist is a set of songs. Sets give us a tidy way to talk about groups, what's in both groups, what's in either and what's left out.

Your Lecture 13 uses sets to describe the kinds of numbers (natural, integer, rational, real) and the inputs a function accepts (Chapter 18). This chapter decodes every symbol, including the scary-looking $\{x \mid 1 \leq x \leq 4\}$.

Goes with **Lecture 13** (types of numbers, sets, domains).

A set is a sock drawer

Open the drawer: a red sock, a blue sock and a green sock. Write the set with **curly brackets** (called **braces**) and commas:



$$S = \{\text{red},\ \text{blue},\ \text{green}\}$$

S is the set's name; each sock is an element

Two rules make a set different from a list:

$$\{\text{red}, \text{blue}, \text{green}\} = \{\text{green}, \text{red}, \text{blue}\}$$

1. order doesn't matter (the drawer isn't sorted)

$$\{\text{red}, \text{red}, \text{blue}\} = \{\text{red}, \text{blue}\}$$

2. repeats don't count ("is red in there?" yes)

Is it in the drawer? The symbol $\in$ (a curly E, for "element") means "is in"; with a slash, $\notin$ means "is not in":

$\text{blue} \in S$ *"blue is in S" (true)*

$\text{pink} \notin S$ *"pink is not in S"*

Sets can hold numbers too: $A = \{2, 4, 6, 8\}$ is the set of even numbers from 2 to 8, and $3 \notin A$. The number of elements is written $n(A)$; here $n(A) = 4$.

Two playlists: in both, in either, only in one

You and a friend each have a playlist. Songs are numbered 1 to 9 in the whole music library. The whole library is called the **universal set** ξ ("ksee", a Greek letter) or U: everything we're allowed to talk about.

$A = \{1, 2, 3, 4\}$ $B = \{2, 3, 7, 9\}$ $\xi = \{1, 2, \ldots, 9\}$

my playlist *friend's playlist*

$A \cap B = \{2, 3\}$

intersection, "A and B"

songs on BOTH playlists. ∩ looks like a bridge (or an upside-down cup) sitting on both.

$A \cup B = \{1, 2, 3, 4, 7, 9\}$

union, "A or B"

merge the playlists, no duplicates. ∪ looks like a cup that holds everything.

$A - B = \{1, 4\}$

difference, "A minus B"

songs only on mine: take away anything your friend also has.

$A' = \{5, 6, 7, 8, 9\}$

complement, "not A"

everything in the library that's NOT on my playlist. (Some books write A with a bar over it.)

Memory hook: ∪ is a **U** for **U**nion; ∩ is the other one. "Or" in maths always includes "both" (song 2 is in A or B, it's in both, so it counts).

RULE: the four set operations

$A \cap B$: elements in A **and** in B. $A \cup B$: elements in A **or** B (or both), each listed once.

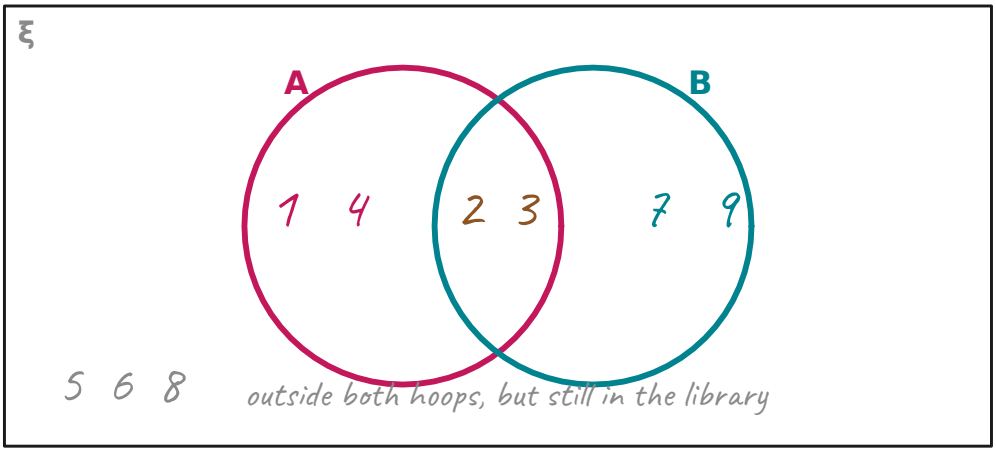
A': elements of ξ that are **not** in A. Counting: $n(A') = n(\xi) - n(A)$.

$A - B$: elements of A that are not in B. Same thing as $A \cap B'$ ("in A and not in B").

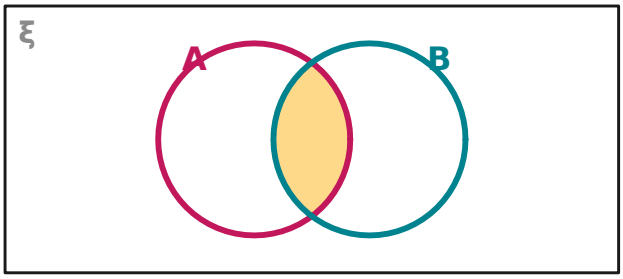
$$n(A') = n(\xi) - n(A) = 9 - 4 = 5$$ ← *matches A' = {5, 6, 7, 8, 9}*

Hula hoops on the floor: Venn diagrams

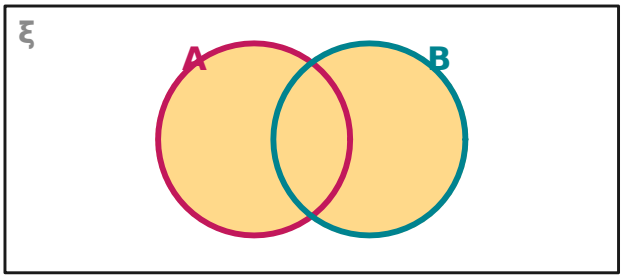
Lay two hula hoops on the floor so they overlap, inside a rectangle (the whole library). Put each song where it belongs. Songs on both playlists stand in the overlap. This picture is a **Venn diagram**:



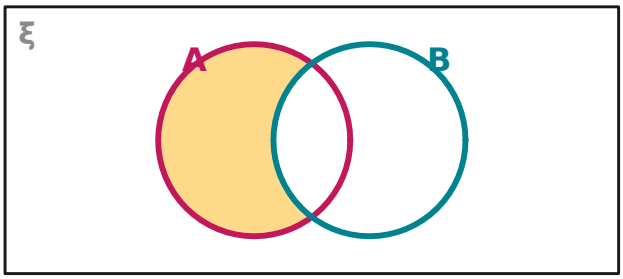
Each operation from the last page is a **region** of the floor. Shade it and read it off:



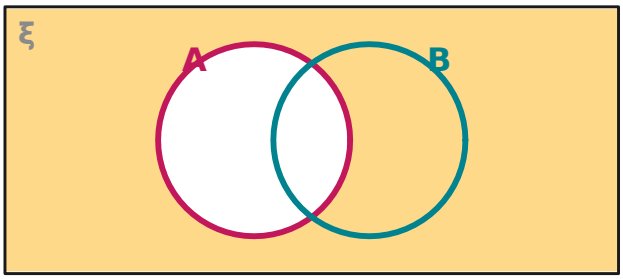
$A \cap B$ the overlap only



$A \cup B$ anything inside a hoop



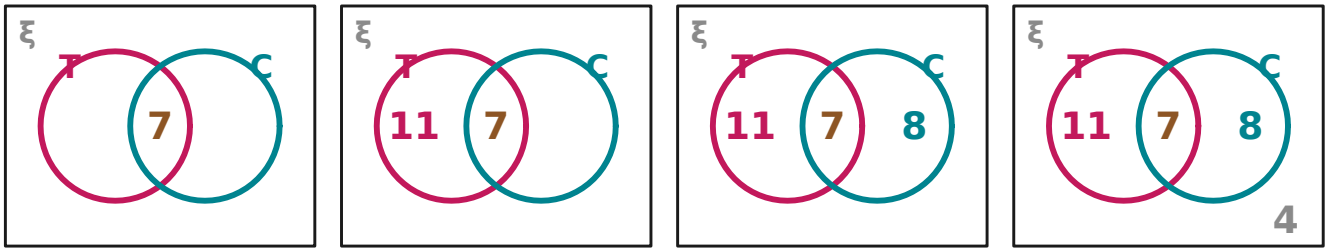
$A - B$ A's hoop, minus the overlap



A' everything outside A

Counting with a Venn diagram: fill it from the middle

A survey of **30** students: **18** drink tea, **15** drink coffee, **7** drink both. How many drink neither? The trick: **always start with the overlap**, then work outwards.



1. both: 7 goes in the middle

2. tea only: 18 - 7 = 11

3. coffee only: 15 - 7 = 8

4. neither: 30 - 11 - 7 - 8 = 4

Check: 11 + 7 + 8 + 4 = 30 students ✓ *(T = tea, C = coffee)*

The classic mistake: 18 + 15 = 33 is more than 30 students! That's because the 7 who drink both got counted twice, once with tea and once with coffee. The fix, as a rule:

RULE: counting a union (inclusion-exclusion)

$n(A \cup B) = n(A) + n(B) - n(A \cap B)$: add the two sets, then take off the overlap you counted twice.

Everyone is somewhere: $n(\xi) = n(A \cup B) + n\big((A \cup B)'\big)$, i.e. total = at least one + neither.

Any three of $n(A)$, $n(B)$, $n(A \cap B)$, $n(A \cup B)$ give you the fourth.

$$n(T \cup C) = 18 + 15 - 7 = 26$$ ← *drink at least one; 30 - 26 = 4 drink neither*

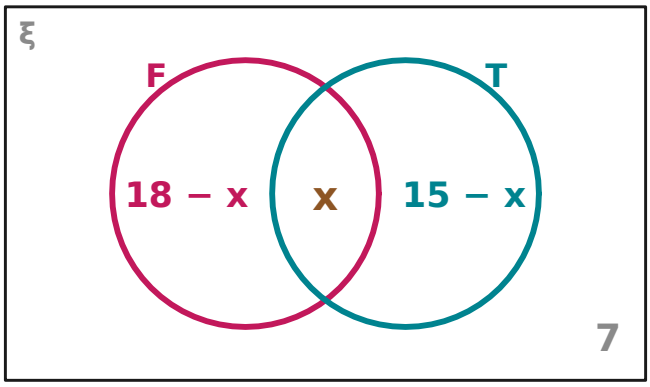
When the overlap is the unknown: call it x

Often you're told the totals and asked for the overlap. You can't start in the middle, so put an **x** there and let the algebra do it.

PROCEDURE: Venn diagram with an unknown

1. Let x be the number in the region you don't know (usually the overlap).
2. Fill every other region in terms of x: A only $= n(A) - x$, B only $= n(B) - x$.
3. Put the "neither" number outside both circles.
4. All regions add up to the total: write that equation and solve for x.
5. Check: put x back in. No region may come out negative.

Example. 30 students: 18 play football, 15 play tennis, 7 play neither. How many play both?



$$(18 - x) + x + (15 - x) + 7 = 30$$

$$40 - x = 30$$

$$x = 10$$

check: 8 + 10 + 5 + 7 = 30 ✓

Same answer from the rule. At least one sport: $30 - 7 = 23$. Then

$$23 = 18 + 15 - n(F \cap T) \;\Rightarrow\; n(F \cap T) = 33 - 23 = 10$$

Finding a set's size instead. $n(A \cup B) = 50$, $n(A) = 32$, $n(A \cap B) = 12$. Find $n(B)$:

$$50 = 32 + n(B) - 12 \;\Rightarrow\; n(B) = 50 - 20 = 30$$

Three sets: same idea, fill from the middle out

With three circles there are **8 regions**: the middle (all three), three "exactly two" regions, three "only" regions, and outside. The method is the same as with two: start in the middle.

PROCEDURE: filling a three-set Venn diagram

1. Middle: write $n(A \cap B \cap C)$.
2. Each pair: $n(A \cap B)$ includes the middle, so "exactly A and B" $= n(A \cap B) -$ middle. Do all three pairs.
3. Each "only": $n(A)$ minus the three numbers already inside circle A.
4. Outside: total minus everything inside the circles.
5. Check: all 8 regions add to the total; none is negative.

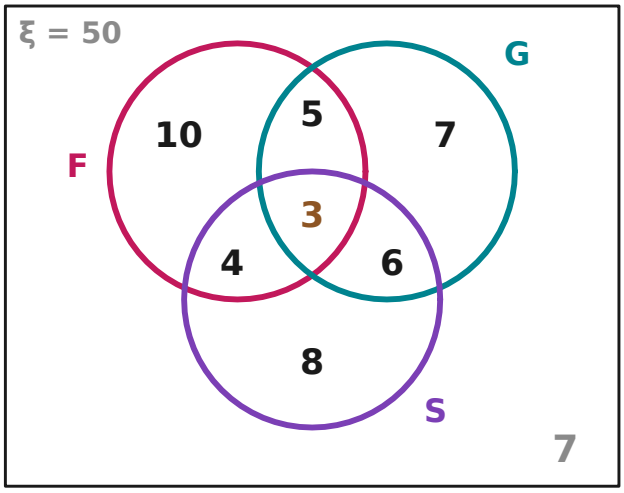
RULE: counting a union of three sets

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$$

Adding the three sets counts the pairs twice, so take them off; that removes the middle three times, so add it back once.

Read carefully: "8 study French and German" normally includes anyone who also studies Spanish. "Only French and German" does not.

Example. 50 students: 22 study French, 21 German, 21 Spanish; 8 study French and German, 7 French and Spanish, 9 German and Spanish; 3 study all three. How many study none? How many study French only?



all three $= 3$

$F \cap G$ only $= 8 - 3 = 5$

$F \cap S$ only $= 7 - 3 = 4$

$G \cap S$ only $= 9 - 3 = 6$

F only $= 22 - 5 - 4 - 3 = 10$

G only $= 21 - 5 - 6 - 3 = 7$

S only $= 21 - 4 - 6 - 3 = 8$

none $= 50 - 43 = 7$

43 = 3 + 5 + 4 + 6 + 10 + 7 + 8, everyone inside

Same answer from the rule:

$$n(F \cup G \cup S) = 22 + 21 + 21 - 8 - 7 - 9 + 3 = 43$$

so 50 - 43 = 7 study none

Complements of unions and intersections (De Morgan)

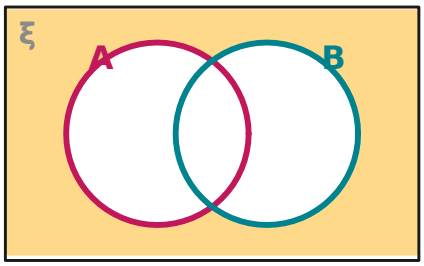
"Not in A or B" is easy to misread. Shade it and you'll see it's the region outside both hoops, which is the same as "not in A **and** not in B".

RULE: De Morgan's laws for sets

$(A \cup B)' = A' \cap B'$: outside the union = outside A and outside B.

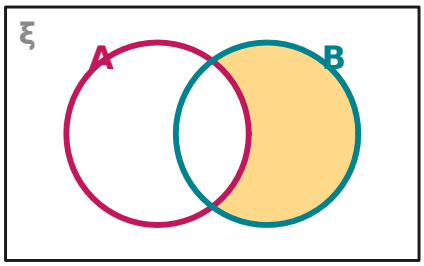
$(A \cap B)' = A' \cup B'$: outside the overlap = outside A or outside B.

Taking the complement swaps $\cup$ and $\cap$ and puts a dash on each set.



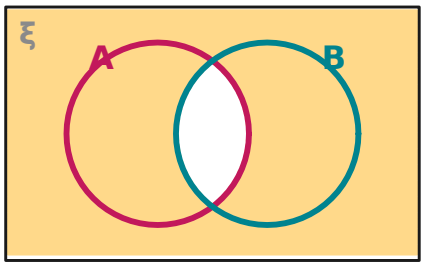
$(A \cup B)'$

outside both



$A' \cap B$

B only



$(A \cap B)'$

all but the overlap

Check on the playlists (ξ = {1, ..., 9}, A = {1, 2, 3, 4}, B = {2, 3, 7, 9}):

$$(A \cup B)' = \xi - \{1,2,3,4,7,9\} = \{5,6,8\}$$

$$A' \cap B' = \{5,6,7,8,9\} \cap \{1,4,5,6,8\} = \{5,6,8\}$$

the same set

$$A' \cap B = \{7,9\}$$ ← *in B, not in A: same as B - A*

✗ Common mistakes

✗ $(A \cup B)' = A' \cup B'$ ✓ $(A \cup B)' = A' \cap B'$

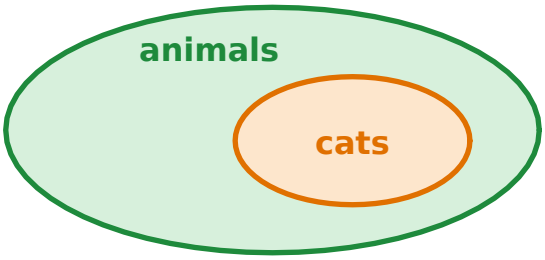
the symbol flips when the dash goes inside

✗ $A' \cap B = (A \cap B)'$ ✓ $A' \cap B = B - A$

only B's own part, not everything outside the overlap

Subsets and the empty set

Every cat is an animal, so the set of cats sits **entirely inside** the set of animals. We say cats are a **subset** of animals, and draw one hoop inside another:



$$\text{Cats} \subset \text{Animals}$$

$\subset$ reads "is a (proper) subset of". It looks like a C for "contained in".

The empty set has nothing in it at all, written $\{\}$ or $\varnothing$. Your lecture's example: the set of fish that drive BMWs. It counts as a subset of **every** set, because it has no element that could fail to be inside.

RULE: subsets

$A \subseteq B$: every element of A is also in B (A may equal B). Every set is $\subseteq$ itself.

$A \subset B$ (**proper** subset): $A \subseteq B$ and $A \neq B$, so B has at least one extra element.

$\varnothing$ is a subset of every set.

A set with n elements has 2^n subsets (each element is either in or out), and $2^n - 1$ proper subsets.

Example. The subsets of $\{a, b, c\}$:

$$\varnothing,\ \{a\}, \{b\}, \{c\},\ \{a, b\}, \{a, c\}, \{b, c\},\ \{a, b, c\}$$

$2^3 = 8$ subsets, $\quad 8 - 1 = 7$ proper ← *the last one is the set itself*

Decoding set-builder notation

Listing works for small sets. For big ones, maths describes the **rule for getting in**, like a club's entry rules. This is from your Lecture 13 slides:

$$\{\ x \mid 1 \leq x \leq 4\ ,\ x \in \mathbb{Z}\ \}$$

the set of all — x — numbers x — such that — $1 \leq x \leq 4$ — x is between 1 and 4, 1 and 4 allowed — $x \in \mathbb{Z}$ — and x is a whole number

$= \{1, 2, 3, 4\}$

Why the "x ∈ ℤ" part matters (it's missing on the slide): without it, the rule $1 \leq x \leq 4$ also lets in 2.5, 3.14 and every other number in between, which is an endless set, not {1, 2, 3, 4}.

Intervals. When you do want every number between two ends, there's a shorter way. The square bracket means the end is **included**; the round bracket means it's **not**:

$[-1, 1] = \{x \mid -1 \leq x \leq 1\}$

$(0, 3) = \{x \mid 0 < x < 3\}$

$[0, 3)$



filled dot = end included, hollow dot = end left out

RULE: interval notation

$[a, b] = \{x \mid a \leq x \leq b\}$ (both ends in); $(a, b) = \{x \mid a < x < b\}$ (both ends out); $[a, b)$ and $(a, b]$ mix the two.

No upper end: $[a, \infty) = \{x \mid x \geq a\}$, $(a, \infty) = \{x \mid x > a\}$. No lower end: $(-\infty, b] = \{x \mid x \leq b\}$.

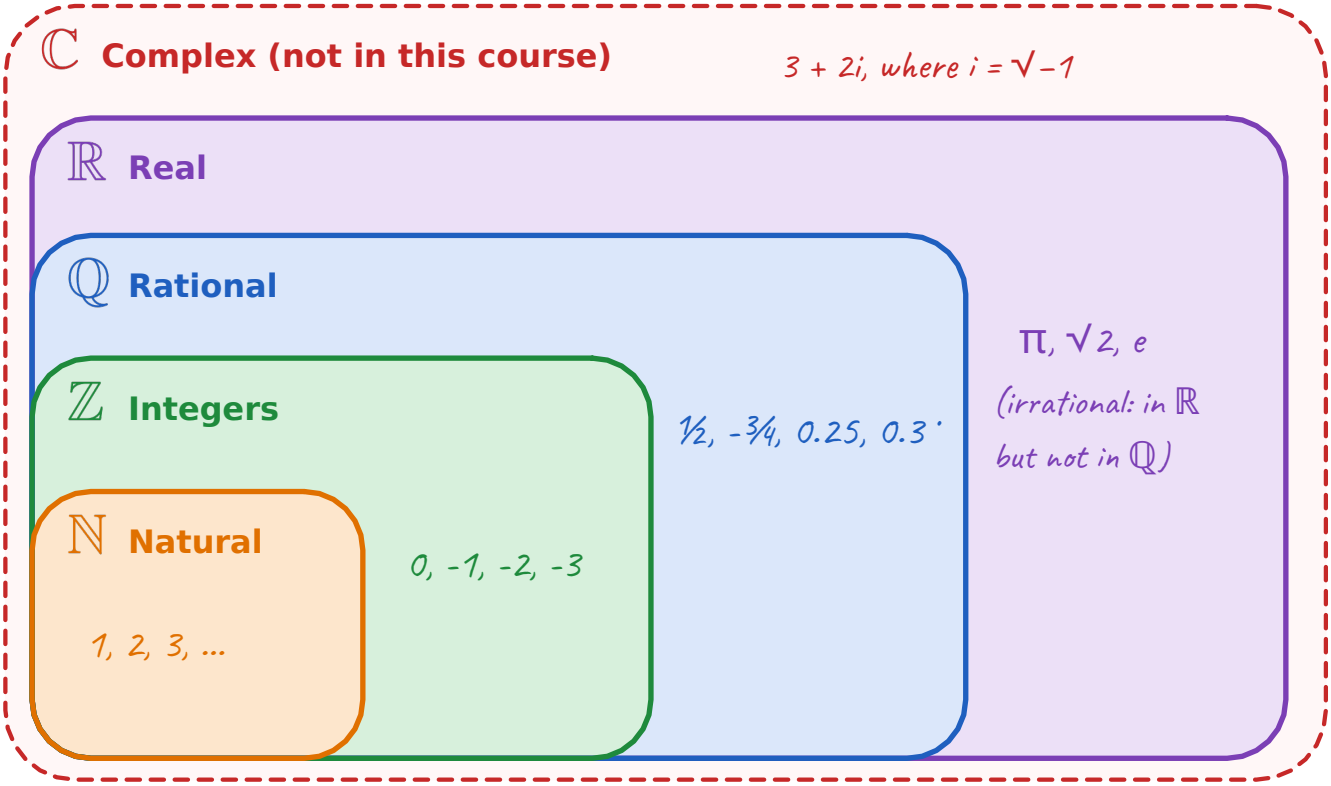
∞ is not a number you can reach, so it always takes a **round** bracket.

Intervals are sets of real numbers: $[0, 3)$ contains 2.9 and 0, but not 3.

$[0, \infty) = \{x \mid x \geq 0\}$ ← zero and every positive number

Nesting dolls: the families of numbers

Numbers come in families, and each family fits inside a bigger one, like Russian nesting dolls. Each has a fancy double-struck letter (from your Lecture 13):



ℕ natural: the counting numbers 1, 2, 3, ... (some books include 0).

ℤ integers: whole numbers, negatives and zero too: ..., -2, -1, 0, 1, 2, ... (Z is from the German "Zahlen", numbers.)

ℚ rational: any fraction of two integers ("ratio-nal"; Q for quotient): ½, 6/1, 3/7, -0.25. Their decimals either stop or repeat (Chapter 3).

ℝ real: every point on the number line, including numbers whose decimals never stop and never repeat: π = 3.14159..., √2 = 1.41421...; these are called irrational.

ℂ complex: the dashed outer doll. No real number squares to give a negative (2^2 and $(-2)^2$ are both +4), so mathematicians invented a new number **i** with $i^2 = -1$. Numbers like 3 + 2i are not on the number line at all, which is why your Lecture 13 defines the reals as "any number that could be used as a distance on a line (so excludes complex numbers)". You won't need them in this course; they turn up later in electronics and waves.

So $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$: each doll fits in the next. The slides write the irrationals as $\mathbb{R} - \mathbb{Q}$ (real minus rational), which is set difference from earlier in this chapter.

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Membership	$x \in A$, $x \notin A$; order and repeats don't matter; $n(A)$ = number of elements
Intersection, union	$A \cap B$: in both; $A \cup B$: in either (or both)
Complement	$n(A') = n(\xi) - n(A)$ A' = everything in ξ not in A
Difference	$A - B = A \cap B'$: in A and not in B
Counting a union	$n(A \cup B) = n(A) + n(B) - n(A \cap B)$ don't count the overlap twice
Venn with an unknown	put x in the unknown region, fill the rest in terms of x, add all regions to the total, solve
Three sets	$n(A \cup B \cup C) = \text{singles} - \text{pairs} + n(A \cap B \cap C)$ singles $= n(A) + n(B) + n(C)$; pairs $= n(A \cap B) + n(A \cap C) + n(B \cap C)$. Venn: fill from the middle out
De Morgan	$(A \cup B)' = A' \cap B'$ and $(A \cap B)' = A' \cup B'$
Subsets	$A \subseteq B$: may be equal; $A \subset B$: proper (B has more); $\varnothing$ is a subset of every set
Number of subsets	$\text{number of subsets} = 2^n$ n elements; $2^n - 1$ proper subsets
Set-builder	$\{x \mid \text{rule}\}$: the set of all x such that the rule holds
Intervals	$[a, b]$, (a, b), $[a, \infty)$: square = end included, round = end left out; ∞ always round
Number families	$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$; irrationals $= \mathbb{R} - \mathbb{Q}$

Hula hoops: your turn

ξ = {1, 2, …, 10}, A = {1, 2, 3, 4, 5}, B = {2, 4, 6, 8, 10}. Answers in grey.

Reading sets

1. Is 6 ∈ A?
2. List A ∩ B
3. List A ∪ B
4. List A - B
5. List B'
6. Find n(A ∪ B)

1. no 2. {2, 4} 3. {1, 2, 3, 4, 5, 6, 8, 10} 4. {1, 3, 5} 5. {1, 3, 5, 7, 9} 6. 8

Venn counting

7. 40 people: 25 have a cat, 20 a dog, 12 both. How many have neither?
8. How many have a cat but no dog?

7. 40 - (13 + 12 + 8) = 7 8. 13

Decoding

9. Write {x | x ∈ ℤ, -2 ≤ x < 3} as a list
10. Which family is the smallest that contains -5?
11. Is 0.75 rational? Is √2?
12. Draw [2, 5) on a number line

9. {-2, -1, 0, 1, 2} 10. ℤ 11. yes (3/4); no 12. filled dot at 2, hollow at 5

Set rules

13. List (A ∪ B)′
14. List A′ ∩ B
15. List (A ∩ B)′ and A′ ∪ B′. Are they equal?
16. How many subsets does {a, b, c, d} have?
17. Is A ⊆ A? Is A ⊂ A?
18. Write {x | x ≥ 4} in interval notation

13. {7, 9} 14. {6, 8, 10} 15. both {1, 3, 5, 6, 7, 8, 9, 10}; yes (De Morgan) 16. $2^4 = 16$ 17. yes; no (a proper subset must be smaller) 18. [4, ∞)

Find the unknown

19. 50 students: 28 play football, 22 play tennis, 9 play neither. How many play both?

20. $n(A \cup B) = 40$, $n(A) = 25$, $n(B) = 21$. Find $n(A \cap B)$

21. $n(A \cup B) = 35$, $n(A) = 20$, $n(A \cap B) = 8$. Find $n(B)$

22. $n(\xi) = 60$ and $n(A') = 42$. Find $n(A)$

23. 32 students: 20 study French, 15 Spanish, 4 neither. How many study both?

24. A set has 64 subsets. How many elements does it have?

19. $(28 - x) + x + (22 - x) + 9 = 50$, so $59 - x = 50$, $x = 9$ 20. $25 + 21 - 40 = 6$ 21. $35 = 20 + n(B) - 8$, $n(B) = 23$ 22. $60 - 42 = 18$ 23. $39 - x = 32$, $x = 7$ 24. $2^n = 64$, $n = 6$

Three sets

25. 100 people: 45 like A, 40 like B, 35 like C; 15 like A and B, 12 A and C, 10 B and C; 5 like all three. How many like none?

26. In the same survey, how many like A only?

25. $45 + 40 + 35 - 15 - 12 - 10 + 5 = 88$ like at least one, so $100 - 88 = 12$ 26. $45 - (15 - 5) - (12 - 5) - 5 = 23$